

[DC] Personality and Affective States in Virtual Reality: A Multi-Study Program on Awe, Acute Stress, and Trust

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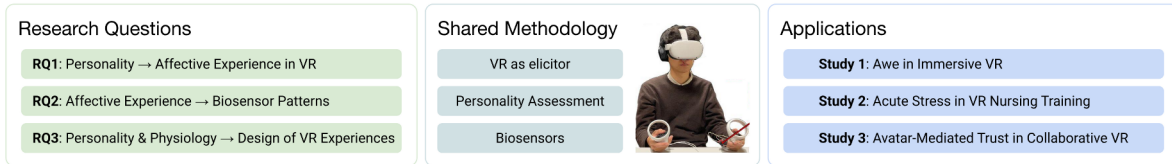


Figure 1: Overview of the Doctoral Thesis' Workflow: Research Questions, Methodology and Applications

ABSTRACT

This doctoral research investigates how personality traits shape affective experiences and physiological responses in Virtual Reality (VR). While VR is widely used to elicit emotions, induce stress, and support social interaction, individual differences are often treated as noise rather than a central focus. Building on the Big Five model, this cumulative thesis examines personality-affect relationships across three complementary studies that share a common methodological core: immersive VR or interactive 3D environments, brief personality assessment, and biosensors. The first, already completed, study explores awe in VR across diverse scenes and cultures, revealing strong effects of scene type and presence but a more complex pattern for personality and physiology than expected. The second study examines acute stress in a VR nursing training scenario, asking how personality moderates stress responses, performance, and acceptance of VR compared to traditional skills lab training. The third study focuses on avatar-mediated trust in collaborative VR, investigating how personality traits influence trust in virtual partners and associated physiological responses. Together, these studies uncover personality-specific affective and physiological patterns in VR and derive design implications for awe experiences, training systems, and avatar design.

Index Terms: Virtual Reality, Personality, Human-Computer Interaction, Affective Computing.

1 INTRODUCTION

Virtual Reality (VR) makes it possible to study affective states in controlled, immersive environments. From self-transcendent emotions such as awe, to task-related stress in training scenarios, to reactions toward avatars, VR allows researchers to present repeatable, immersive situations while tightly controlling sensory input and interaction. This makes VR an attractive laboratory for affective computing, human-computer interaction, and psychological research [5].

People do not experience VR in a uniform way. Personality traits, often conceptualized via the Big Five model [1] and measured efficiently with brief instruments such as the Ten Item Personality Inventory (TIPI) [3], capture stable interindividual differences in tendencies such as Openness to Experience, Neuroticism, Extraversion,

Agreeableness, and Conscientiousness. Prior work suggests that these traits modulate how strongly individuals experience emotions, stress, and cognitive load, and how they respond to complex environments. Some people appear more sensitive to acute stressors [4], while others report stronger or more frequent positive affects [2]. However, VR studies often treat personality as a control variable rather than a central focus.

Affective states themselves span multiple constructs, including discrete emotions (e.g., awe, fear), task-related experiences (e.g., workload, acute stress), and social-relational responses (e.g., trust, comfort, acceptance). In VR, these states can be induced through carefully designed scenes and tasks and assessed via self-report and biosensors (e.g., skin conductance, respiration, heart rate), although links between subjective and physiological responses are often complex and context-dependent.

This doctoral research links personality, affective states, VR, and biosensor-based measurement in a cumulative program. The overall goal is to understand how personality traits modulate VR-induced affective experiences and their physiological manifestations in three contexts: (1) awe in immersive environments, (2) acute stress in VR nursing training, and (3) avatar-mediated trust in collaborative VR. Each context is addressed in a dedicated paper using common personality and physiology measures. Across these papers, I will address three central research questions (RQ):

RQ1 How do personality traits influence the intensity and quality of affective experiences in VR (e.g., awe, acute stress, avatar-mediated trust)?

RQ2 How consistently are these experiences reflected in biosensor data across different VR applications?

RQ3 How can insights about personality-specific affective and physiological responses inform the design of VR experiences, such as awe-eliciting content, training scenarios that calibrate stress, or avatar designs that foster trust?

2 RESEARCH PLAN

My dissertation is structured as a cumulative thesis consisting of three main studies, each realized as (or planned to become) a separate publication. Together they address how personality traits shape affective experiences and physiological responses in VR across three different contexts: awe, acute stress in nursing training, and avatar-mediated trust. All three studies share a brief personality measure (TIPI), VR as eliciting stimulus, and biosensors to capture autonomic responses. This common methodology enables direct comparison and synthesis in the final thesis.

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2.1 The Awe-Some Spectrum: Self-Reported Awe Varies by Eliciting Scenery and Presence in Virtual Reality, and the User's Nationality

The first study [7] investigates awe in VR across different scene types and nationality backgrounds. Participants from Jordan, Japan and Germany experienced ten immersive VR scenes (e.g., natural landscapes, human-made structures, abstract spaces) designed to elicit varying degrees of awe. After each scene, they reported their level of awe and presence, while biosensors recorded skin conductance and respiration throughout the experience.

The results confirm that VR can reliably elicit awe and that ratings vary systematically with scene content and nationality background. However, physiological measures did not show a simple, robust relationship with awe ratings. Regarding personality, Openness to Experience, which previous work associated with stronger awe [6], did not correlate with self-reported awe in this study, but did show a relationship with skin conductance responses. This unexpected pattern raises questions about how robust personality-affect links remain when moving from traditional stimuli (e.g., films) to immersive VR, and how consistently they appear in physiological data.

2.2 Stressors in VR Nursing Training: Personality, Acute Stress, and Learning Outcome (ongoing)

The second study focuses on acute stress in a VR nursing training scenario. Nursing students will practice clinical procedures in both a conventional skills lab and an immersive VR environment. The training tasks will be designed to include realistic stressors, such as time pressure, multitasking demands, and simulated patient interactions, while maintaining pedagogical relevance.

We will combine subjective measures of stress, usability, and training effect with biosensor data (skin conductance, heart rate, respiration), performance metrics (task success, errors, completion times), and TIPI-based personality scores. Analytically, we will use mixed-effects models to examine how personality moderates stress responses in VR versus traditional training, and how these responses relate to learning outcomes and the perceived usefulness of VR.

This study extends RQ1 and RQ2 beyond positive self-transcendent states to a task-focused context. It will test, for example, whether individuals high in Neuroticism show stronger physiological stress in VR, whether highly conscientious students perform better under higher workload, and whether openness is again linked to distinct physiological patterns. This shared framework enables direct comparison of personality-physiology couplings across positive and stressful states.

2.3 Avatar-Mediated Trust: Personality in Collaborative VR (planned)

The third study examines avatar-mediated trust as affective state in collaborative VR settings. Participants will interact with another avatar in a short cooperative tasks, which require them to rely on their virtual partner. The avatar's appearance will be systematically manipulated along a realism-stylization dimension via facial proportion changes. Here, social and self-related affective responses will be assessed through trust self-reports (e.g., willingness to follow advice or to rely on information), behavioral indicators (e.g., choices), possible willingness to disclose sensitive information, and physiology (skin conductance, respiration, heart rate).

Personality traits are expected to modulate trust. For example, I will test whether Agreeableness and Extraversion predict higher baseline trust and social comfort with avatars, whether Neuroticism is associated with reduced trust and heightened arousal when interacting with ambiguous or less friendly-looking avatars, and whether Openness supports stronger acceptance of stylized or unconventional avatars. Comparing these patterns with those from the awe and nursing studies will show whether personality-physiology relationships

are specific to particular VR contexts or generalize across positive, stressful, and social states, thereby addressing RQ1–RQ3.

3 CONCLUSION

Taken together, the three experiments contribute to how personality shapes affective experiences and physiological responses in diverse VR contexts, using personality and biosensor measures to move beyond isolated case studies. Scientifically, I expect to contribute:

- a cross-context characterization of when and how personality traits modulate VR-induced affective states
- evidence on the consistency or specificity of personality-physiology links across positive, stressful, and social affect
- design-relevant insights for awe-eliciting experiences, VR training scenarios, and avatar design, particularly regarding how to support users with different personality profiles

Methodologically, all studies use the same pipeline (VR, brief personality assessment, multimodal biosensors), which provides consistency across the three application areas.

At the Doctoral Consortium, I hope to receive feedback on both the conceptual framing and the concrete study designs. In particular, I would like to discuss the following questions:

- **Theoretical integration and scope:** How can I best position a cumulative thesis that spans awe, acute stress, and avatar-mediated trust as one coherent contribution to “personality and affect in VR,” without appearing too broad or fragmented?
- **Measurement and modeling of personality × physiology:** Given the mixed findings on personality-affect relationships in the awe study (e.g., Openness not correlating with self-reported awe but relating to skin conductance), how should I refine my personality measurement and analysis strategy in the upcoming studies?
- **Design trade-offs and comparability across studies:** In the nursing and avatar projects, how can I balance ecological validity (realistic tasks, meaningful decisions, plausible avatars) with experimental control, so that personality effects and physiological patterns remain comparable across all three studies?

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